

Elderly Care Monitoring App

Project Overview

The Elderly Care Monitoring App is a facility-focused web platform for monitoring residents, managing incidents, tracking medications, and supporting coordination between administrators, caregivers, and family members.

Technology Stack

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS
- Lucide React (icons)
- Recharts (analytics charts)

Backend

- Node.js
- Express.js
- Zod (request validation)
- JWT (authentication)
- bcryptjs (password hashing)

Database

- SQLite (local persistent database)

Security and Access Control

- Role-based access control:
- Admin
- Caregiver
- Relative/Family
- Protected API routes using Bearer tokens
- Password policy and first-login password change
- Session revoke/logout-all support

Integration and Utilities

- UniSMS integration for SMS alert testing
- Local backup/restore utilities for SQLite

Core Features

1. Authentication and User Management

- Secure login with JWT
- Admin bootstrap flow for first production admin
- User CRUD for admin (create, update role, activate/deactivate, reset password, delete)
- Profile security modal (change password and revoke sessions)

2. Resident Management

- Create, view, edit, and archive resident profiles
- Personal information management:
- Date of birth
- Computed age
- Admitted date from actual creation timestamp

- Photo upload and crop support
- Caregiver assignment per resident

3. Real-Time Monitoring Dashboard

- Separate dashboards by role (admin, caregiver, relative)
- Live resident monitoring cards
- Active incident status visibility
- Caregiver task tracking (pending and completed)

4. Alert and Incident Handling

- Alert simulation for demo use:
- Fall detected
- Sleep anomaly
- Unusual pulse
- Alert acknowledgment and resolution workflow
- Incident history page with status filtering
- Role-aware modal prompts for critical incidents

5. Medication Management

- Add medication schedules with controlled resident selection
- Medication status updates (Pending/Given)
- Event tracking for medication actions (assigned, pending, given)
- Unified visibility for admin/caregiver/family based on permissions
- Medication logs with actor and timestamp metadata

6. Caregiver Task Management

- Admin can assign care tasks to caregivers
- Tasks can be marked done/undone
- Task logs preserve completed activity history
- Dashboard now separates pending items from completed logs

7. Family Portal Features

- Family accounts linked to specific resident IDs
- Family resident selector for multi-linked access
- Read-only monitoring access for selected residents:
- Incident history
- Medication logs
- Health records

8. Reports and Auditability

- Resident-level report view with health, incidents, and medication context
- Audit logs for critical system actions:
- User changes
- Resident changes
- Device changes
- Medication updates
- Alert status changes

9. Device and System Management

- Device CRUD and assignment support
- Battery/status tracking
- System management panel for role assignments and linking workflows

10. Demo and Deployment Readiness

- Local dev setup with combined or split frontend/backend execution
- LAN and ngrok sharing options for presentation/demo

- Backup and restore runbook for operational reliability

Current Project Strengths

- End-to-end role-based workflows
- Practical demo simulation tools for thesis presentation
- Strong traceability through audit and medication event logs
- Stable resident ID mapping and family data scoping
- PH timezone-consistent date/time display across UI